FERNS



THE MOST ADVANCED of all the non-flowering, spore-bearing plants are the ferns and their relatives, known as the Pteridophytes. There are about 12,000 species of Pteridophyte, of which about 10,400 are ferns. The others include horsetails and club mosses. Pteridophytes are vascular plants, that is, plants whose stems contain tissues that transport water and food around the plant internally. They flourish best in warm, damp environments, but also grow where it is cool or dry.

Male fern
Ferns similar to this male fern, so-called because of its vigorous growth, are found in woods all around the world. This species has stiff, bright green fronds. Each blade is divided into "leaves" called pinnae (singular: pinna), each of which is further divided into pinnules.

Pinna and pinnules

A full-grown frond may reach as much as 150 cm (5 ft) long.

The frond continues to lengthen as the lower parts unfurl.

Young frond

A developing male fern plant

Stipe is the "stalk" of the fern.

Rhizome

Root

Ferns

A typical fern plant has underground stems, or rhizomes, from which grow roots, and leaves called fronds. Upright rhizomes produce a fern with a short radiating crown of fronds, while long horizontal rhizomes produce a spreading fern. Ferns grow in a variety of places, but all have a two-stage life-cycle. The gametophyte is a small, short-lived plantlet that produces sex cells. After fertilization, a female sex cell grows into a sporophyte, which is the fern plant.

1 Frond buds develop on the rhizome. Each bud produces just one frond. It takes up to three years for a bud to develop and a frond to start growing.

2 A frond can grow rapidly because all the cells of the stalk and leaflets are fully formed, though very small. They just have to expand.

3 A male fern produces tall fronds, each on a scaly stipe. Fertile fronds are the last to unfurl, so that the spores are released in the summer.

Epiphytic ferns
An epiphytic fern grows on the bough or trunk of a tree. It takes no nourishment from its host, but obtains moisture and minerals from rain and debris that become trapped among its roots.

Fronds hang clear of the branch.





Radiating fronds

Base of dead fronds

Tree ferns
Tree ferns have woody, fibrous trunks topped with a crown of fronds. They are found in all climates, most frequently in the tropics and sub-tropics. The tallest species reaches 20 m (65 ft).

Water ferns
Some ferns are aquatic. They either root into mud in fresh water, or float free. This *Azolla* species floats. Its tiny roots dangle in the water below a mat of fronds.



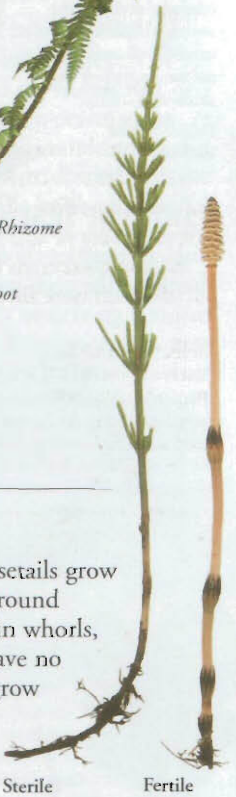
Club mosses

These small plants grow on damp ground or on rainforest trees. Their creeping stems are covered with tiny leaves arranged in a spiral. Spores are carried in modified leaves on fertile stems.



Horsetails

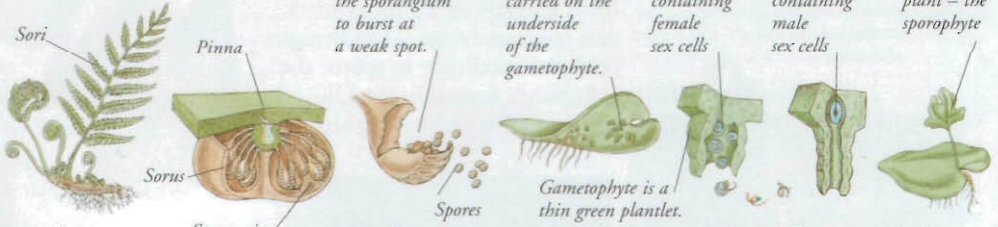
The stiff upright stems of horsetails grow in dense patches from underground stems. Branches are arranged in whorls, although fertile stems often have no branches. Tiny brown leaves grow in rings around the stems and branches. Spores are borne in cone-like structures on the tip of fertile stems.



Sterile stem

Fertile stem

The life-cycle of a fern



Fertile fronds have sori (singular: sorus), usually on the lower surface of each pinna.

Within each sorus are clusters of sporangia, which contain the spores.

Sixty-four spores develop inside each sporangium. The sporangium then bursts, releasing all the spores.

A spore landing on damp soil germinates into a gametophyte. This bears male and female sex organs.

Male sex cells swim in a film of soil moisture to the female sex cells and then fertilize them.

The first female sex cell to be fertilized grows into a fern plant.

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